Optimizing CubeSat Bus Design for Versatile Payload Integration: Standardization through Additive Manufacturing

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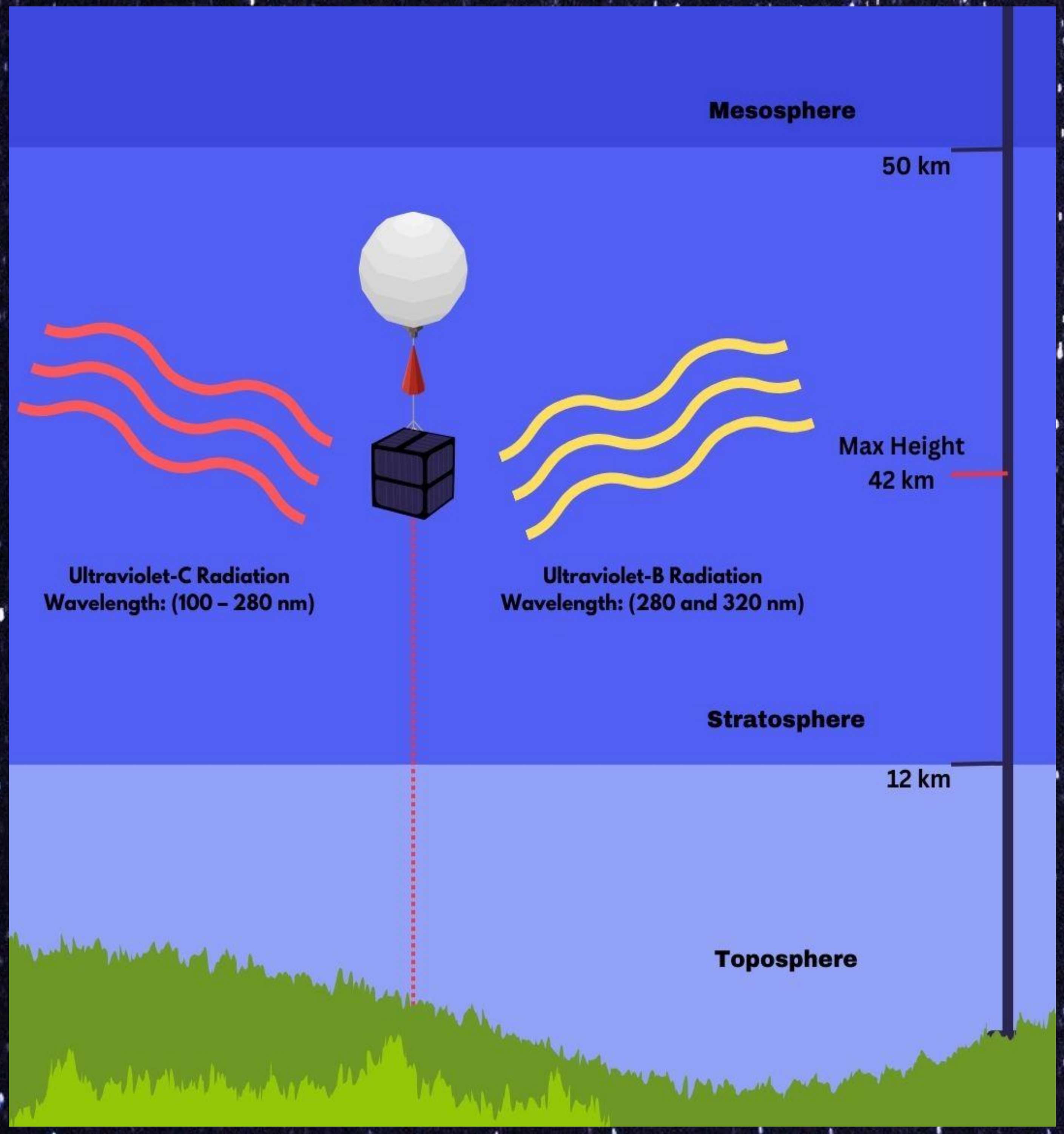
Overview

LEARNSat is a UGA SSRL's latest mission focused on making CubeSat technology accessible to K12 Institutions. CubeSats are used in a wide variety of research applications but assembling and deploying them are a costly endeavour. In order to create a way for high schools to emulate the CubeSat assembly process and conduct meaningful research without having exorbitant deployment costs, we at SSRL are developing the LEARNSat Kits. LEARNSat Kits will consist of the Satellite Bus, hardware foundation that provides CubeSats with power, communications, thermal stability, and other services a payload needs to function. LEARNSats will then be launched using weather balloons, reducing the cost of launch drastically. This research will focus on creating the standardized structural 2U frame that will house the rest of the satellite bus.

Design and Development

- **Additive manufacturing** was chosen as the method of production due to its ability to produce complex and lightweight structures, Design Flexibility, cost-effectiveness and standardization potential.
- Gyroid infill pattern was chosen for its superior strength-to-weight ratio and reduced mass. A study by the University of Defense, Czech Republic, found that gyroid infill prints 23.73% faster than full honeycomb while offering 6.81% greater average strength and 12.59% higher tensile strength.
- Liquid Bed Fusion is the chosen printing methodology for its superior strength-to-weight ratio, minimal warping, and high surface quality.



Environmental Factors

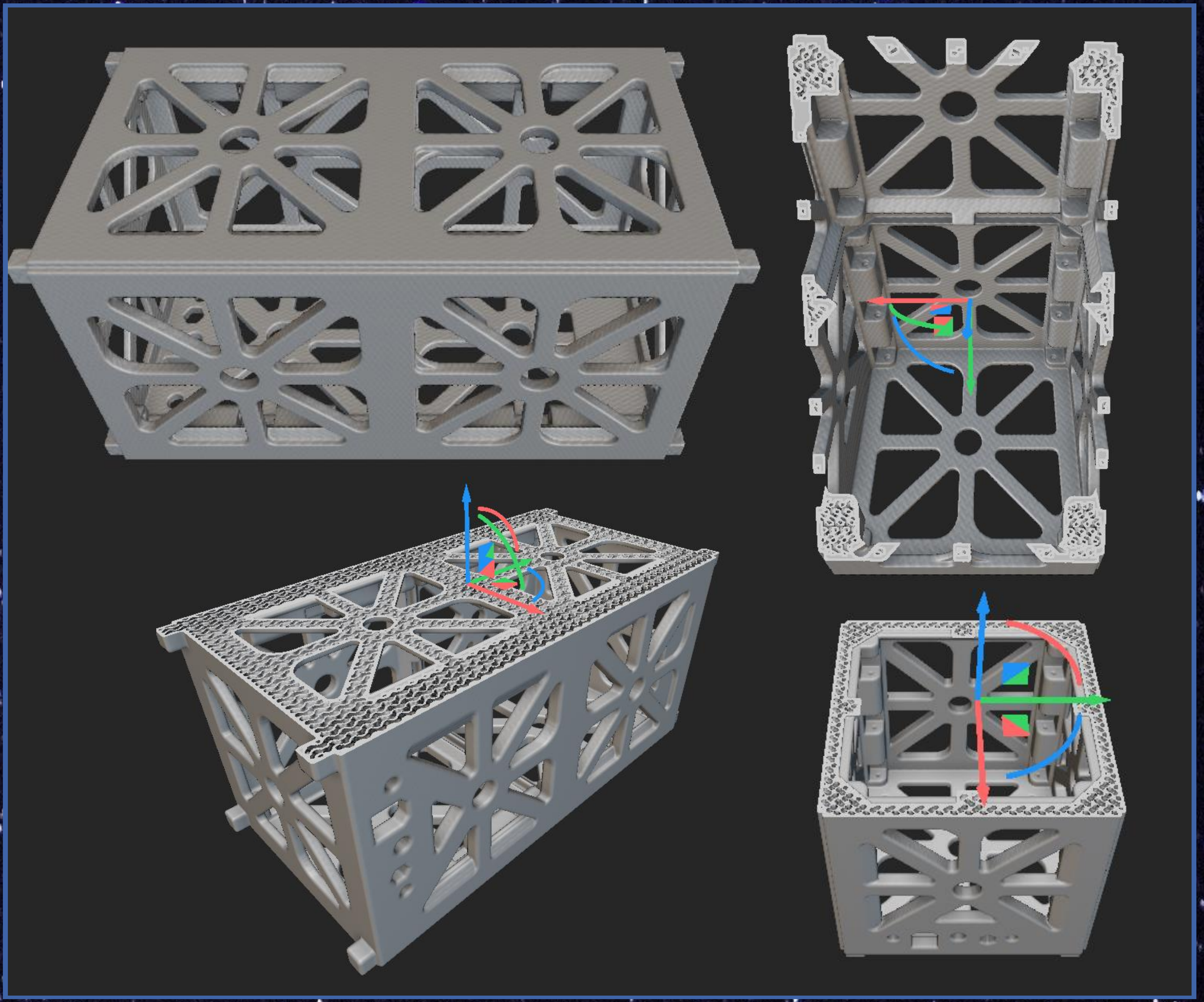
- Stark Temperature changes $\Delta T = 76^\circ$ C
- Pressure Difference: $\Delta P = 100.669$ kPa
- UV Radiation

Methodology

To develop the external shell, 3 separate applications were used:

1. **Autodesk Inventor** (for developing CAD Bodies)
2. **nTop** (for latticing and gyroid infill
3. **Thermal Desktop** (for testing and simulations).

Meshing → Latticing → Material Reduction → Simulations and Testing



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